4. Introducing random variables

We use the usual notations

Ω - sample space, $\mathcal{F}$ - events, $\mathbb{P}$ - probability measure.

<u>def</u>: We call $X: \Omega \mapsto \mathbb{R}$ a <u>random variable</u> if for any interval I in $\mathbb{R}$: $X^{-1}(I) \in \mathcal{F}$.

<u>Example</u>: Toss a coin three times

$\Omega = \{(H,H,H),(H,H,T),\dots\}$, $\mathcal{F} = P(\Omega)$, X: = number of heads

$X \in \{0,1,2,3\}$

$$\mathbb{P}(\underbrace{\{X=0\}}_{=\{\omega\in\Omega:\, X(\omega)=0\}}) = \mathbb{P}(\{(H,H,H)\}) = {1/2}^3 = \underline{\underline{1/8}}$$

$$\mathbb{P}(X=1) = \underline{\underline{\frac{3}{8}}}, \quad \mathbb{P}(X=2) = \underline{\underline{\frac{3}{8}}}, \quad \mathbb{P}(X=3) = \underline{\underline{\frac{1}{8}}}$$

<u>def</u>: We call a random variable <u>discrete</u> if it can take only finitely or infinitely but countable many values.

<u>def</u>: The <u>distribution</u> of a discrete random variable $X: \Omega \mapsto \mathbb{R}$ is a $p: \mathbb{R} \mapsto [0,1]$ such that

$$p(y) = \mathbb{P}(X=y) = \mathbb{P}(\{\omega\in\Omega:\, X(\omega)=y\})$$

If X is a discrete random variable with values $\{x_i\}_{i\in I}$, then

- $\forall i \in I:\ p(x_i) \geq 0$ ← weight of x_i
- $\sum_{i\in I} p(x_i) = \sum_{i\in I} \mathbb{P}(X=x_i) = \mathbb{P}(\bigcup_{i\in I}\{X=x_i\}) = \mathbb{P}(\Omega) = 1$
- $\forall x\in\mathbb{R}:\ p(x) = \begin{cases} \mathbb{P}(X=x), & x\in\{x_i\}_{i\in I} \\ 0, & \text{otherwise} \end{cases}$

<u>Example</u>: Let X be a discrete random variable with values in $\mathbb{N}$ such that

$p(i) = c\cdot\frac{\lambda^i}{i!}$ where $\lambda > 0$, for $i\in\mathbb{N}$

What values can c take?

$$1 = \sum_{i=0}^{\infty} p(i) = \sum_{i=0}^{\infty} c\cdot\frac{\lambda^i}{i!} = c\cdot e^{+\lambda} \Rightarrow \underline{\underline{c = e^{-\lambda}}}$$

What is the probability of the event $\{X=0\}$?

$$\mathbb{P}(X=0) = e^{-\lambda}\cdot\frac{\lambda^0}{0!} = \underline{\underline{e^{-\lambda}}}$$

What is the probability that X is bigger than 2?

$$\mathbb{P}(X>2)=\mathbb{P}\left(\bigcup_{i=3}^{\infty}\{X=i\}\right)=\sum_{i=3}^{\infty}\mathbb{P}(X=i)=\underline{\underline{\sum_{i=3}^{\infty}e^{-\lambda}\frac{\lambda^i}{i!}}},$$

or we can take the complementer event

$$\mathbb{P}(X>2)=1-\mathbb{P}(X\leq 2)=1-\mathbb{P}(X=0)-\mathbb{P}(X=1)-\mathbb{P}(X=2)=\underline{\underline{1-e^{-\lambda}-\lambda e^{-\lambda}-\frac{\lambda^2}{2}e^{-\lambda}}}.$$

I. Parameters of distributions

def. We call x the <u>mode</u> of a discrete random variable X if

$$\forall y\in\mathbb{R}: p(x)\geq p(y).$$

The mode always exists, but not necessarily unique.

def. Let X be a discrete random variable with values $\{x_i\}_{i\in I}$. If $\sum_{i\in I}|x_i|p(x_i)<\infty$, then we say that the <u>expected value</u> of X is

$$\mathbb{E}[X]=\sum_{i\in I}x_i\,p(x_i).$$

If $\sum_{i\in I}|x_i|p(x_i)=\infty$, then we say that the expected value does not exist.

<u>Example</u>: Toss a dice. What is the expected value of the result?

$\Omega=\{1,2,3,4,5,6\}$, $X(i)=i$

$$\mathbb{E}[X]=1\frac{1}{6}+2\cdot\frac{1}{6}+3\frac{1}{6}+4\frac{1}{6}+5\frac{1}{6}+6\frac{1}{6}=\underline{\underline{\frac{7}{2}}}$$

<u>Note</u>: $\mathbb{P}(X=7/2)=0$

def. Let $(\Omega,\mathcal{F},\mathbb{P})$ be a probability space. Let $E\in\mathcal{F}$ be an event. We call $\mathbb{1}_E$ the indicator of E if

$$\mathbb{1}_E(\omega)=\begin{cases}1, & \omega\in E\\ 0, & \omega\notin E.\end{cases}$$

Prop. $\mathbb{E}[\mathbb{1}_E]=\mathbb{P}(E)$

<u>proof</u>: $$\mathbb{E}[\mathbb{1}_E]=0\cdot\mathbb{P}(\mathbb{1}_E=0)+1\cdot\mathbb{P}(\underbrace{\mathbb{1}_E=1}_{=\{\omega\in\Omega:\ \omega\in E\}})=\mathbb{P}(E)\qquad\square$$

Prop. Let $g:\mathbb{R}\mapsto\mathbb{R}$ and let X be a discrete random variable with values $\{x_i\}_{i\in I}$. Then for $Y=g(X)$ we have

$$\mathbb{E}[g(X)]=\mathbb{E}[Y]=\sum_{i\in I}g(i)\,\mathbb{P}(X=x_i).$$

proof: Let $\{y_j\}_{j\in J} = \{g(x_i)\}_{i\in I}$

$$\mathbb{E}[g(X)] = \mathbb{E}[Y] = \sum_{j\in J} y_j \, \mathbb{P}(Y=y_j) = \sum_{j\in J} y_j \, \mathbb{P}(\underbrace{g(X)=y_j}_{=\bigcup_{i\in I: g(x_i)=y_j}\{X=x_i\}}) = \sum_{j\in J} y_j \sum_{\substack{i\in I\\ g(x_i)=y_j}} \mathbb{P}(X=x_i) =$$

$$= \sum_{j\in J} \sum_{\substack{i\in I\\ g(x_i)=y_j}} \overbrace{y_j}^{g(x_i)} \mathbb{P}(X=x_i) = \sum_{j\in J} \sum_{\substack{i\in I\\ g(x_i)=y_j}} g(x_i)\,\mathbb{P}(X=x_i) = \sum_{i\in I} g(x_i)\,\mathbb{P}(X=x_i). \quad \square$$

Note: $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$!!

Example: Let X be a random variable such that

$\mathbb{P}(X=-1)=0{,}2$, $\mathbb{P}(X=0)=0{,}5$, $\mathbb{P}(X=1)=0{,}3$.

$\mathbb{E}[X] = (-1)\cdot 0{,}2 + 0\cdot 0{,}5 + 1\cdot 0{,}3 = \underline{0{,}1}$

$$\mathbb{E}[X^2] = \begin{cases} 0\cdot \mathbb{P}(X^2=0) + 1\cdot \mathbb{P}(\underbrace{X^2=1}_{=\{X=-1\}\cup\{X=1\}}) = 0\cdot 0{,}5 + 1\cdot 0{,}5 = \underline{\underline{0{,}5}} \neq (0{,}1)^2 = (\mathbb{E}[X])^2 \\ (-1)^2 0{,}2 + 0^2 0{,}5 + 1^2 0{,}3 = \underline{\underline{0{,}5}} \end{cases}$$

Prop. (Linearity of expected value) Let X be a discrete random variable, then $\forall a,b\in\mathbb{R}$: $\mathbb{E}[aX+b] = a\,\mathbb{E}[X]+b$.

proof: Let $\{x_i\}_{i\in I}$ be the possible values of X.

$$\mathbb{E}[aX+b] = \sum_{i\in I}(ax_i+b)\,\mathbb{P}(X=x_i) = a\underbrace{\sum_{i\in I} x_i \mathbb{P}(X=x_i)}_{=\mathbb{E}[X]} + b\underbrace{\sum_{i\in I}\mathbb{P}(X=x_i)}_{=1} = a\,\mathbb{E}[X]+b. \quad \square$$

Prop. Let X and Y be discrete random variables, then
$$\mathbb{E}[X+Y] = \mathbb{E}[X]+\mathbb{E}[Y].$$

proof: Let $Z=X+Y$, then the possible values of Z are

$\{x_i+y_j\}_{i\in I, j\in J} =: \{z_k\}_{k\in K}$, where $\{x_i\}_{i\in I}$, $\{y_j\}_{j\in J}$ are the possible values of X and Y respectively.

$$\mathbb{E}[Z] = \sum_{k\in K} z_k \mathbb{P}(Z=z_k) = \sum_{k\in K} z_k \mathbb{P}\Big(\bigcup_{\substack{i\in I, j\in J\\ x_i+y_j=z_k}} \{X=x_i\}\cap\{Y=y_j\}\Big) =$$

$$= \sum_{k\in K} z_k \sum_{\substack{i\in I, j\in J\\ x_i+y_j=z_k}} \mathbb{P}(\{X=x_i\}\cap\{Y=y_j\}) = \sum_{k\in K} \sum_{\substack{i\in I, j\in J\\ x_i+y_j=z_k}} (x_i+y_j)\,\mathbb{P}(\{X=x_i\}\cap\{Y=y_j\}) =$$

$$= \sum_{i\in I, j\in J} x_i \mathbb{P}(\{X=x_i\}\cap\{Y=y_j\}) + \sum_{i\in I, j\in J} y_j \mathbb{P}(\{X=x_i\}\cap\{Y=y_j\}) =$$

$$= \sum_{i\in I} x_i \, \mathbb{P}\Big(\big(\underbrace{\bigcup_{j\in J}\{Y=y_j\}}_{=\Omega}\big)\cap\{X=x_i\}\Big) + \sum_{j\in J} y_j \, \mathbb{P}\Big(\big(\underbrace{\bigcup_{i\in I}\{X=x_i\}}_{=\Omega}\big)\cap\{Y=y_j\}\Big) =$$

$$= \sum_{i\in I} x_i \, \mathbb{P}(X=x_i) + \sum_{j\in J} y_j \, \mathbb{P}(Y=y_j) = \mathbb{E}[X] + \mathbb{E}[Y].$$

def. Let X be a discrete random variable. Then we call the variance of X the quantity $\mathbb{E}[(X-\mathbb{E}[X])^2] =: \mathbb{D}^2[X] =: Var(X)$.

The standard deviation of X is $\mathbb{D}[X] := \sqrt{\mathbb{D}^2[X]}$

Prop. $\mathbb{D}^2[X] \geq 0$, $\mathbb{D}[X] \geq 0$

Prop. $\mathbb{D}^2[X] = \mathbb{E}[X^2] - \mathbb{E}^2[X]$

proof: $\mathbb{D}^2[X] = \mathbb{E}[(X-\mathbb{E}[X])^2] = \mathbb{E}[X^2 - 2X\mathbb{E}[X] + \mathbb{E}^2[X]] =$

$$= \mathbb{E}[X^2] - \mathbb{E}[2X\mathbb{E}[X]] + \mathbb{E}[\mathbb{E}^2[X]] = \mathbb{E}[X^2] - \cancel{2}\,\mathbb{E}^2[X] + \cancel{\mathbb{E}^2[X]} = \mathbb{E}[X^2] - \mathbb{E}^2[X]. \quad \square$$

Cor. $\mathbb{E}[X^2] \geq \mathbb{E}^2[X]$

Example: Toss a dice. What is the variance of the result?

We've already seen that $\mathbb{E}[X] = 7/2 \Rightarrow \mathbb{E}^2[X] = 49/4$

$$\mathbb{E}[X^2] = 1^2\frac{1}{6} + 2^2\frac{1}{6} + 3^2\frac{1}{6} + 4^2\frac{1}{6} + 5^2\frac{1}{6} + 6^2\frac{1}{6} = \frac{91}{6}$$

$$\Rightarrow \mathbb{D}^2[X] = \mathbb{E}[X^2] - \mathbb{E}^2[X] = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}$$

Prop. $\mathbb{D}^2[aX+b] = a^2\mathbb{D}^2[X]$ and $\mathbb{D}[aX+b] = |a|\cdot\mathbb{D}[X]$

proof: $\mathbb{D}^2[aX+b] = \mathbb{E}[(aX+b-\mathbb{E}[aX+b])^2] = \mathbb{E}[(aX+\cancel{b}-a\mathbb{E}[X]-\cancel{b})^2] =$

$$= \mathbb{E}[a^2(X-\mathbb{E}[X])^2] = a^2\mathbb{D}^2[X].$$

$$\mathbb{D}[aX+b] = \sqrt{\mathbb{D}^2[aX+b]} = |a|\cdot\mathbb{D}[X] \quad \square$$

II. Discrete distributions

Example: Suppose that an experiment can have two possible outcomes, success or failure, and that the probability of success is p.

Let $X := \mathbb{1}_{success}$ be a random variable.

$p(0) = \mathbb{P}(X=0) = 1-p$

$p(1) = \mathbb{P}(X=1) = p$

We call X a Bernoulli random variable.

Note: $X \sim Ber(p)$

$\mathbb{E}[X] = p$, $\mathbb{D}^2[X] = \underbrace{\mathbb{E}[X^2]}_{=1^2 p + 0^2(1-p) = p = \mathbb{E}[X]} - \mathbb{E}^2[X] = p - p^2 = p(1-p)$

Example: We repeat the same experiment n times independently.
Let Y be the number of successful experiments.

$Y \in \{0, 1, \dots, n\}$ ← possible values Y might take

$$p(k) = \mathbb{P}(Y=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

← the probability that the others fail

$\binom{n}{k}$: choose the k successful experiments

p^k: the probability of them succeeding

Note: We call Y a binomial random variable with parameters n and p.
We use the notation $Y \sim Bin(n,p)$.

Prop: Let $Y \sim Bin(n,p)$, then $\mathbb{E}[Y] = np$

proof:

$$\mathbb{E}[Y] = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k} = \sum_{k=1}^{n} k \cdot \frac{n!}{\underset{(k-1)!}{k!}(n-k)!} p^k (1-p)^{n-k} = \sum_{k=1}^{n} \frac{(n-1)!}{(k-1)!(n-k)!} n p^k (1-p)^{n-k} =$$

$$= \sum_{k=1}^{n} \binom{n-1}{k-1} n p^k (1-p)^{n-k} = n \sum_{k=0}^{n-1} \binom{n-1}{k} p^{k+1} (1-p)^{n-k-1} = np \underbrace{\sum_{k=0}^{n-1} \binom{n-1}{k} p^k (1-p)^{n-k-1}}_{\text{binomial theorem} = (p+1-p)^{n-1} = 1} = \underline{\underline{np}} \qquad \square$$

Prop: Let Y be a binomial random variable with parameters n and p. ($Y \sim Bin(n,p)$)
Then $\mathbb{D}^2[Y] = np(1-p)$.

proof:

$$\mathbb{D}^2[Y] = \mathbb{E}[Y^2] - \mathbb{E}^2[Y] = \underbrace{\mathbb{E}[Y(Y-1)]}_{\text{we just need to calculate this term.}} + \mathbb{E}[Y] - \mathbb{E}^2[Y]$$

$$\underline{\mathbb{E}[Y(Y-1)]} = \sum_{k=0}^{n} k(k-1)\mathbb{P}(Y=k) = \sum_{k=0}^{n} k(k-1)\binom{n}{k} p^k (1-p)^{n-k} =$$

← the first two terms of the sum are zeros.

$$= \sum_{k=2}^{n} k(k-1) \frac{(n-2)!}{\underset{(k-2)!}{k!}(n-k)!} n(n-1) p^k (1-p)^{n-k} = n(n-1) \sum_{k=2}^{n} \binom{n-2}{k-2} p^k (1-p)^{n-k} =$$

like in the previous proof

$$= n(n-1)p^2 \underbrace{\sum_{k=0}^{n-2} \binom{n-2}{k} p^k (1-p)^{n-k-2}}_{\text{binomial theorem} = 1} = \underline{n(n-1)p^2}$$

$$\Longrightarrow \mathbb{D}^2[Y] = n(n-1)p^2 + np - n^2p^2 =$$

$$= \cancel{n^2p^2} - np^2 + np - \cancel{n^2p^2} = \underline{\underline{np(1-p)}} \qquad \square$$

<u>Prop.</u> Let $Y \sim Bin(n,p)$, then the mode is $\lfloor (n+1)p \rfloor$.
If $(n+1)p$ is an integer, then $(n+1)p$ and $(n+1)p-1$ are both the modes of Y.

<u>proof:</u>

$$\frac{P(Y=k)}{P(Y=k-1)} = \frac{\binom{n}{k} p^k (1-p)^{n-k}}{\binom{n}{k-1} p^{k-1} (1-p)^{n-k+1}} = \frac{\cancel{n!}}{k!(n-k)!} \cdot \frac{(k-1)!(n-k+1)!}{\cancel{n!}} \cdot p \cdot \frac{1}{1-p} = \frac{n-k+1}{k} \frac{p}{1-p}$$

$$\frac{n-k+1}{k} \frac{p}{1-p} \geq 1 \quad \Longleftrightarrow \quad (n+1)p \geq k$$

It implies two things:

- As far as $k \leq (n+1)p$ we have $p(k-1) < p(k) \Rightarrow$ the mode is $\lfloor (n+1)p \rfloor$.
- For $k = (n+1)p$ we have $p(k) = p(k-1) \Rightarrow$ if $(n+1)p$ is an integer, then
  integer
  $(n+1)p$ and $(n+1)p-1$ are both modes. $\square$